

STM32CubeIDE

File Project Run Window Help

main.c main.h main.c main.c main.c .cproject .mxproject .project MX STM32F103C8TX_FLA... MX *Homework.ioc

Homework.ioc - Pinout & Configuration

Pinout & Configuration

Categories A->Z

System Core

- DMA
- GPIO
- IWDG
- NVIC
- RCC
- SYS**
- WWDG

Analog >

Timers >

Connectivity >

Computing >

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Clock Configuration

Software Packs

Project Manager

Pinout

Tools

SYS Mode and Configuration

Mode

Debug Serial Wire

☐ System Wake-Up

Timebase Source SysTick

Pinout view

System view

Configuration

Warning: This peripheral has no parameters to be configured.

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RCC Mode and Configuration

Categories A-Z

System Core

DMA

GPIO

IWDG

NVIC

✓ RCC

✓ SYS

WWDG

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Mode

High Speed Clock (HSE) Crystal/Ceramic ...

Low Speed Clock (LSE) Disable

☐ Master Clock Output

Configuration

Reset Configuration

✓ NVIC Settings

✓ GPIO Settings

✓ Parameter Settings

✓ User Constants

Configure the below parameters :

 Search (Ctrl+F)

System Parameters

VDD voltage (V) 3.3 V

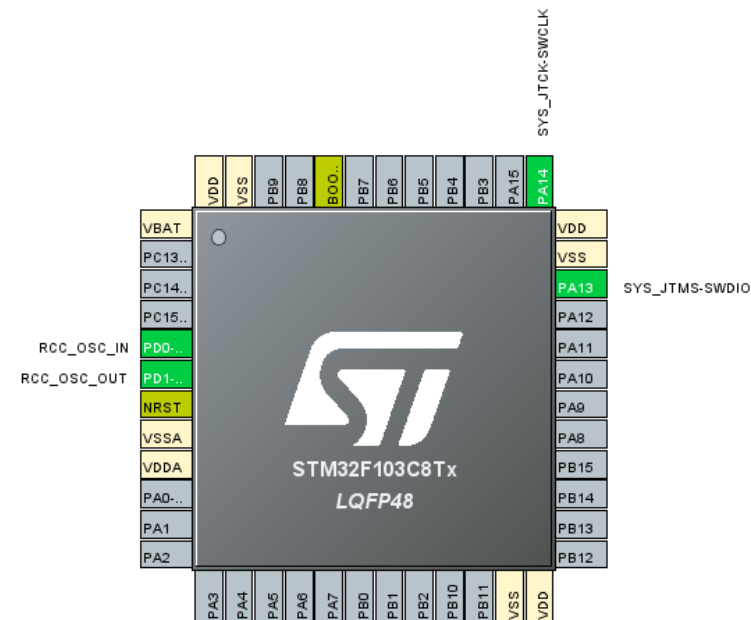
Prefetch Buffer Enabled

Flash Latency... 0 WS (1 CPU cycle)

RCC Parameters

Pinout view

System view



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RCC Mode and Configuration

Mode

- High Speed Clock (HSE) Crystal/Ceramic Reson...
- Low Speed Clock (LSE) Disable
- ☐ Master Clock Output

Configuration

Reset Configuration

- ☒ NVIC Settings
- ☒ GPIO Settings
- ☒ Parameter Settings
- ☒ User Constants

Configure the below parameters :

Search (Ctrl+F)

System Parameters

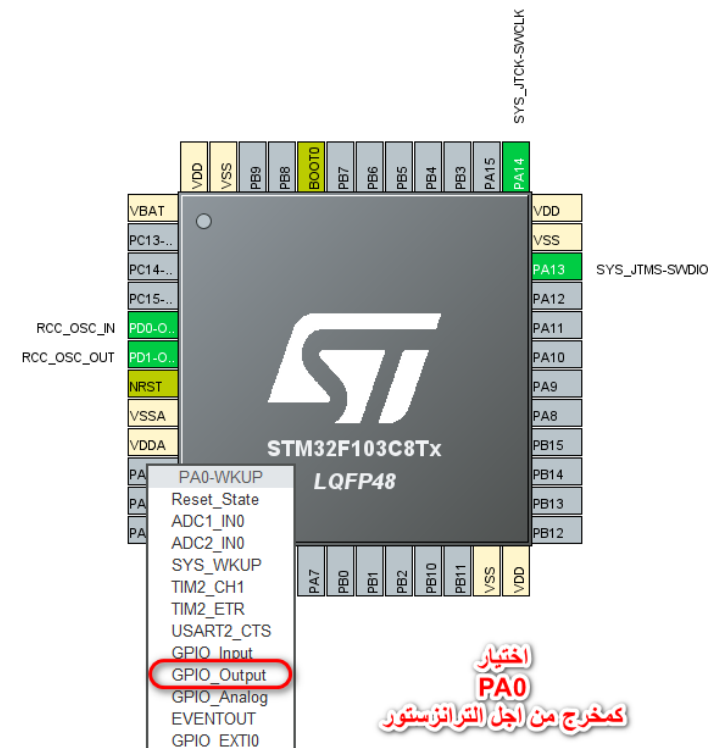
- VDD voltage (V) 3.3 V
- Prefetch Buffer Enabled
- Flash Latency (WS) 2 WS (3 CPU cycle)

RCC Parameters

- HSI Calibration V... 16
- HSE Startup Tim... 100
- LSE Startup Timo... 5000

Pinout view

System view



Configuration Tool - STM32CubeIDE

Project Run Window Help

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Pinout & Configuration

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RCC Mode and Configuration

Mode

High Speed Clock (HSE) Crystal/Ceramic Reson...

Low Speed Clock (LSE) Disable

☐ Master Clock Output

Configuration

Reset Configuration

☒ NVIC Settings ☒ GPIO Settings

☒ Parameter Settings ☒ User Constants

Configure the below parameters :

Search (Ctrl+F)

System Parameters

- VDD voltage (V) 3.3 V
- Prefetch Buffer Enabled
- Flash Latency(WS) 2 WS (3 CPU cycle)

RCC Parameters

- HSI Calibration V... 16

Clock Configuration

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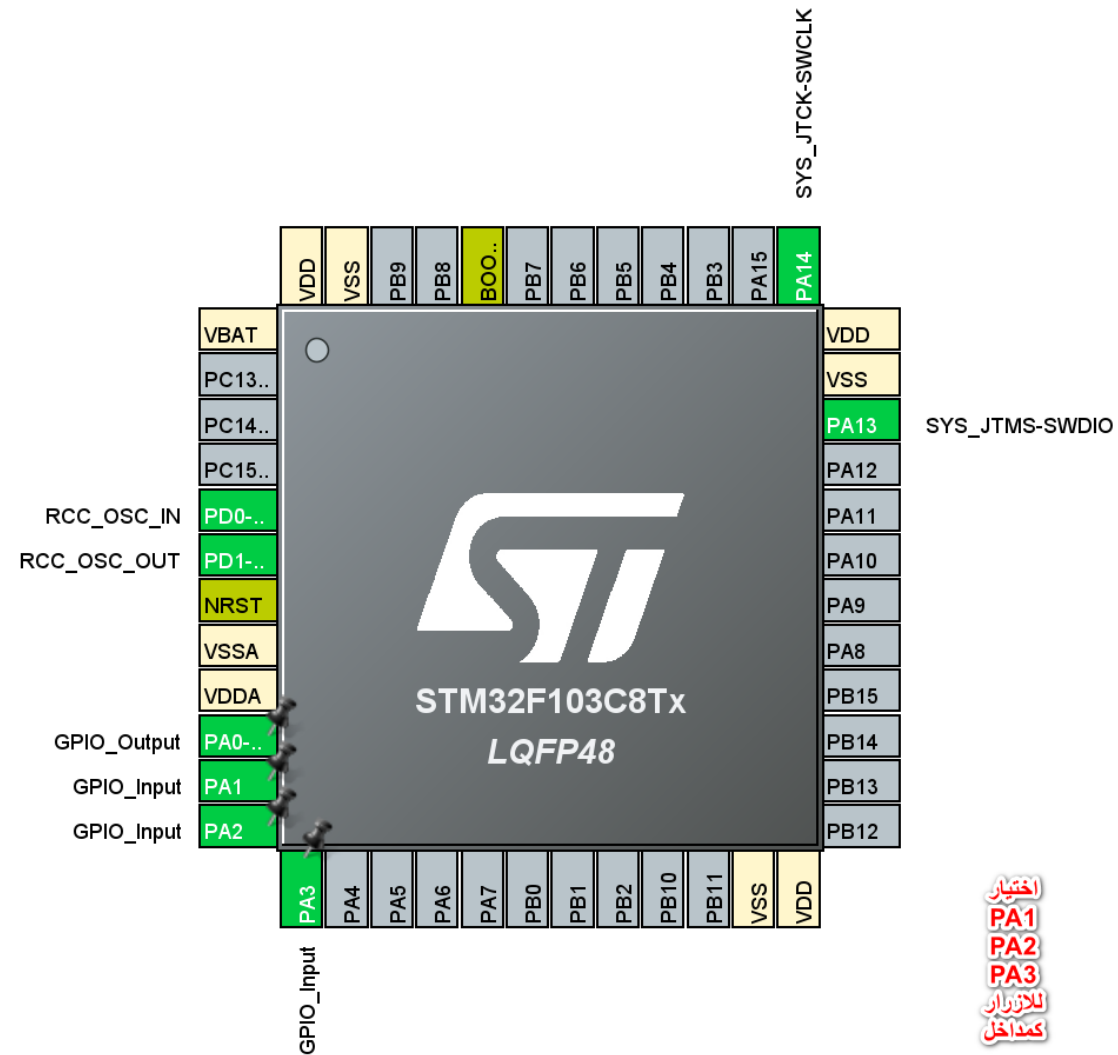
Pinout view System view

Pinout diagram of the STM32F103C8Tx LQFP48 package. The diagram shows the pin connections for the microcontroller. The pins are labeled as follows:

- Top row: VDD, VSS, PB9, PB8, **BOOT0**, PB7, PB6, PB5, PB4, PB3, PA15, PA14, VDD.
- Right side: VSS, PA13, PA12, PA11, PA10, PA9, PA8, PB15, PB14, PB13, PB12.
- Bottom row: PA3, PA4, PA5, PA6, PA7, PB0, PB1, PB2, PB10, PB11, VSS, VDD.
- Left side: VBAT, PC13-..., PC14-..., PC15-..., PD0-O..., PD1-O..., NRST, VSSA, VDDA, PA0-..., PA1, PA2.

Labels for specific pins and functions:

- RCC_OSC_IN: PD0-O...
- RCC_OSC_OUT: PD1-O...
- GPIO_Output: PA0-...
- SYS_JTCK-SWCLK: PA14
- SYS_JTMS-SWDIO: PA13



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main.cmain.hmain.cmain.cmain.c.cproject.mxproject.projectMXSTM32F103C8TX_FLASH.ldMX*Homework.ioc

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CAN

I2C1

I2C2

SPI1

SPI2

USART1

USART2

USART3

USB

Computing>

Middleware>

I2C1 Mode and Configuration

Mode

I2C I2C

Configuration

Reset Configuration

NVIC Settings

DMA Settings

GPIO Settings

Parameter Settings

User Constants

Configure the below parameters :

Search (Ctrl+F)

Master Features

I2C Speed ModeStandard Mode

I2C Clock Speed (Hz)100000

Slave Features

Pinout viewSystem view

اعداد I2C

I2C1_SDAI2C1_SCLSYS_JTCK-SWCLK

VDDVSSPB9PB8BOOT0PB7PB6PB5PB4PB3PA15PA14

VBATPC13-...PC14-...PC15-...PD0-O...PD1-O...NRSTVSSAVDDA

STM32F103C8TxLQFP48

GPIO_OutputGPIO_InputGPIO_Input

PA3PA4PA5PA6PA7PB0PB1PB2PB10PB11VSSVDD

PA13PA12PA11PA10PA9PA8PB15PB14PB13PB12

SYS_JTMS-SWDIO

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TIM3 Mode and Configuration

Mode

Slave Mode Disable

Trigger Source Disable

- ☒ Internal Clock

Channel1 Disable

Channel2 Disable

Channel3 Disable

Channel4 Disable

Combined Channels Disable

- ☐ *XOR activation*

☐ One Pulse Mode

Configuration

✔ NVIC Settings
✔ DMA Settings

Parameter Settings

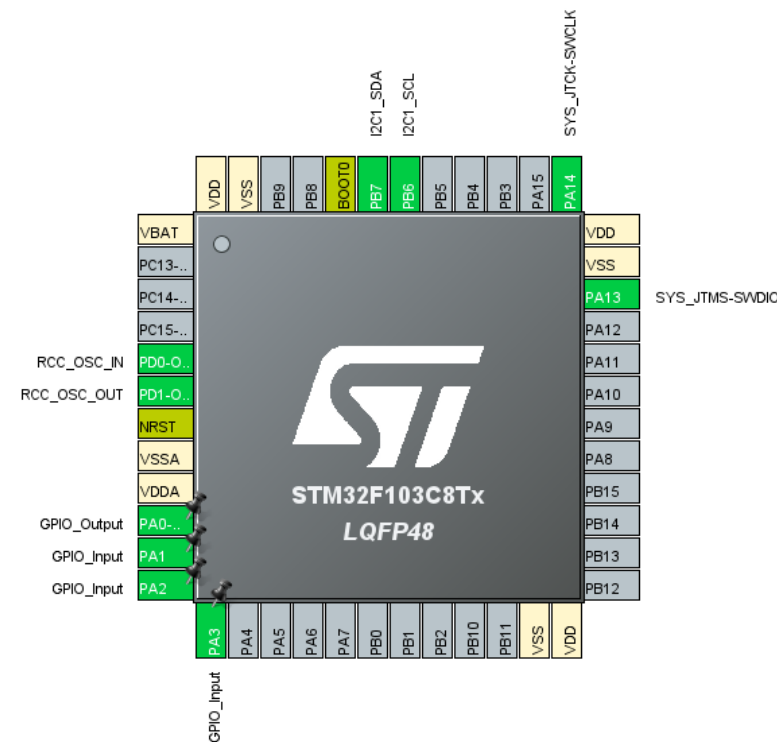
NVIC Interrupt Table	Enabled	Preemption Priority	S
----------------------	---------	---------------------	---

TIM3 global interrupt	☒	0	0
-----------------------	-------------------------------------	---	---

تفعيل المقاطعة

 Pinout view

 System view



Tools

▼ Pinout

TIM3 Mode and Configuration

System view

A-Z



RTC

RTC

TIM1

TIM2

 **TIM3**

 **TIM4**

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Mode

Slave Mode Disable

Trigger Source Disable

- ☒ Internal Clock

Channel1 Disable

Channel2 Disable

Channel3 Disable

Channel4 Disable

Combined Channels Disable

- ☐ *XOR activation*

☐ One Pulse Mode

Configuration

Reset Configuration

✓ NVIC Settings

DMA Settings

✔ Parameter Settings

- ✓ User Constants

Configure the below parameters :

Search (Ctrl+F) ⏪ ⏩

▼ Counter Settings

Prescaler (PSC - 16 bits... 7199

Counter Mode Up

Counter Period (AutoRel... 4999

Internal Clock Division (... No Division

auto-reload preload Disable

er Output (TRGO) Paramet...

Master/Slave Mode (MS... Disable (Trigger input effect not del...

Trigger Event Selection Reset (UG bit from TIMx_EGR)

